

Chapter – 4: Exploring Magnets

- Reshma – coastal town – Kerala – fond – writing stories
- Her grandma – loves – listening to them
- Reshma – new story – grandma's 60th b'day
- Story – ship – carry spices – from Kerala – trade – olden days
- Reshma – aware – those days – sailors – used stars – find dir. – night time
- BUT – her story – situation – sailors – caught in storm – overcast sky – stars not visible
- Reshma – dilemma – how to proceed – how to find dir.
- Researched – library, internet – found out – device – magnetic compass – find directions
- Pencil box, purse – magnets – close them – even duster – BUT – never obs. closely
- Now – Reshma – curious – how do they work
- Magnets – sailors used – naturally occurring – lodestones
- Later – people found – magnets – can be made
- Nowadays – magnets – diff. materials – various shapes – all artificial

Magnetic and Non-Magnetic Materials

- Activity –
 - Collect few items – diff. materials
 - **Predict** – which will stick – note down
 - Hold a magnet – bring near obj. – **obs.** – which actually stick
- Some obj. – attracted to magnet – **magnetic materials**
 - Iron, nickel, cobalt, some of their alloys
- Some obj. – not attracted to magnet – **non-magnetic materials**
 - Plastic, paper, wood

Poles of Magnet

- Activity –
 - Spread – iron fillings – sheet of paper
 - Place a bar magnet – tap the paper – obs. carefully
- Max. iron fillings – stick near the ends – very few – stick in the middle
- These ends – poles of magnet – North and South pole
- Magnet – any shape – max. iron fillings – stick to poles
- Single pole – cannot exist
- Break magnet – smaller pieces – will also have pair of poles

Finding Directions

- Activity –
 - Suspend – bar magnet – with thread – middle of magnet
 - Adjust magnet – balance horizontally
 - Rotate magnet – horizontally – let it rest
 - Mark the position – ends of magnets – join with a line
 - Repeat multiple times – mark the position every time
- Freely suspended magnet – rest – north-south direction – earth – natural magnet

- End of magnet – points north – North-seeking pole – **North Pole**
- End of magnet – points south – South-seeking pole – **South Pole**
- Repeat – iron bar – does not rest – north-south direction
- Only magnets – rest in north-south direction – used – find directions
- Based on this – small device – magnetic compass – invented
- Compass – magnet – shape of needle – rotate freely – indicates – north-south direction
- Place the compass – find directions – let the needle rest – rotate compass – align marked north and south with the needle
- That's how you find directions
- Activity –
 - Collect few materials – cork, iron, sewing needle, bar magnet, glass bowl, water
 - Place – sewing needle – wooden table – keep 1 pole – on the needle – move along its length – reach the needle's end – lift the magnet
 - Bring – same pole to same end of needle – repeat step – 30 to 40 times
 - Bring – iron fillings – near needle – if – iron fillings – attracted – needle converted to magnet
 - Pass the needle – cork – float in water
 - Needle – rests – north-south direction
 - Rotate cork – let it rest – note the direction – always – north-south direction

Attraction and Repulsion between Magnets

- Activity –
 - Take a pair of bar magnets – mark as A and B
 - Place longer side – magnet A – over 5-6 pencils
 - Bring 1 end – magnet B – near the magnet A – make sure – they do not touch – obs.
 - Bring another end – magnet B – near magnet A – same end – obs.
- Unlike poles – 2 magnets – attract each other
- Like poles – 2 magnets – repel each other
- Activity –
 - Take – magnetic compass, bar magnet
 - Place – compass – horizontal surface – let the needle rest
 - Bring north pole – bar magnet – near compass – obs. – needle deflects
 - Repeat – with south pole
- North pole – bar magnet – north pole – compass needle – it moves away
- South pole – bar magnet – north pole – compass needle – it comes closer
- Activity –
 - Repeat above activity
 - Without disturbing – place – piece of wood – between them
 - No effect on deflection
 - Repeat the process – replace – wood with other items – cardboard, thin plastic, thin glass
- No appreciable change in deflection – **conclude** – magnetic effect – act through non-magnetic materials

Fun with Magnets

- After learning – Reshma – excited – set up fun activities – school fair
- Make a garland out of thumb pins and bar magnet
- Take steel balls out of maze by moving magnet below tray
- Pick out steel paper clip out of water without getting wet

- Cars fitted with magnet run away or towards each other

